PRODUCT REQUIREMENTS DOCUMENT

Multimodal E-Commerce Re Classification

Optimizing Sentiment Analysis Using Vision-Language APIs

Document version

1.0

Target platform

Web demo (Streamlit or Gradio) with API backend

Project type

Final-year academic project

1. Executive Summary

Traditional sentiment analysis relies solely on text, which can fail in e-comm where physical product context matters. A review stating “The screen is hug positive for a television but negative for a smartphone. Text alone also cann reported physical defect or determine whether an uploaded image supports

This project proposes a multimodal classification system that uses a vision- such as OpenAI GPT-4o Vision or Gemini 1.5 Pro, to analyze customer revie optional product image together. The system classifies sentiment as **GOOD BAD**, detects visible defects, and assesses whether the image aligns with th

2. System Architecture and Technology Stack

Layer	Technology / responsibility
Frontend / UI	Streamlit or Gradio interface for review text and image upload.
Multimodal engine	Vision-language LLM API (for example, GPT-4o Vision or Gemini 1.5 Pro).
Backend logic	Python 3.10+ for prompt orchestration, API calls, validation, and JSON parsing.
Data processing	Text sanitization, image resizing, Base64 encoding, and structured prompt templatin
Evaluation	Scikit-learn for classification reports, macro F1, and confusion matrices against a lab

3. Target Use Cases

- • **Sentiment classification:** Classify a review as GOOD, NEU using the combined context of its text and product photo.

- - **Defect and claim verification:** Assess whether an uploade visibly supports a complaint, such as a claim that an item is broken.
- - **Visual-text mismatch detection:** Flag a review whose ima unrelated to the text, such as a positive shoe review accompanied by a product image.

4. Functional Requirements

ID	Feature	Description	Priority
FR1	Multimodal input handling	Accept review text of up to 500 words and an optional JPG or PNG image up to 5 MB and image together.	High
FR2	Image preprocessing	Resize images to optimize API payload size and encode them as Base64 when requir provider.	High
FR3	Zero-shot prompt engineering	Supply a structured prompt that instructs the API to act as an e-commerce review cla evaluate text and image evidence.	High
FR4	Three-class sentiment output	Return a validated structured JSON response with sentiment exactly one of GOOD, N BAD.	High
FR5	Visual defect flagging	Include a boolean `defect_detected` field based on visible image evidence; do not in defect.	Medium
FR6	Reasoning extraction	Return a brief `reasoning` string citing relevant textual and visual evidence, when ava	Medium
FR7	Visual-text mismatch flagging	Include a boolean `visual_text_mismatch` field indicating whether the image appea with the product or claims in the review.	Medium

Response contract: `{"sentiment":"GOOD|NEUTRAL|`

`BAD","defect_detected":true,"visual_text_mismatch":false,"` `"Brief evidence-based explanation."}`. Boolean values are example must return values supported by its analysis.

5. Non-Functional Requirements

Requirement	Target	Verification
Classification quality	Macro F1-score greater than 0.85 on a defined multimodal test dataset, with perform against a text-only baseline.	Evaluate on held-out labeled examples; report dataset size, class balance, and per-c
Latency	End-to-end classification response within 3.5 seconds.	Measure elapsed time across representative requests and report observed distributio may vary.
Robustness	Handle missing inputs gracefully. In particular, classify text-only reviews without an i	Test text-only requests, invalid files, and malformed or incomplete API responses.
Cost efficiency	Keep API execution cost below $0.01 per review during testing.	Record provider, model, token/image usage, and contemporaneous pricing for each e

6. Pipeline Workflow

1. **Ingestion:** The user enters review text and optionally uploa image.
2. **Formatting:** The backend validates input limits, sanitizes t image, and encodes it for the selected API.
3. **Inference request:** The backend submits the prompt and a modalities with instructions for structured output.
4. **Vision-language processing:** The model analyzes sentime defects, and image-text consistency, cross-referencing claims such a with visible evidence.
5. **Output parsing:** The backend validates the response field sentiment values, handling invalid responses safely.
6. **Display:** The UI presents the sentiment, defect and mismat concise reasoning.

7. Academic Evaluation and Benchmarking

- **Ablation study:** Compare text-only classification, image-o where labels permit, and multimodal text-plus-image classification.
- **Confusion matrices:** Compare class-level errors and exam

multimodal evidence helps distinguish NEUTRAL from BAD compared only baseline.

- - **Cost-benefit analysis:** Document financial and computati between API-based multimodal inference and a local CNN-plus-LSTM

Evaluation results depend on the dataset, model, prompt, and provider. The latency, and cost targets are project objectives to be measured, not assume

8. Project Deliverables

Deliverable	Description
Source code	Python backend and web UI hosted on GitHub.
Evaluation notebook	Jupyter notebook that evaluates the dataset through the API and generates accuracy classification reports, and graphs.
Live demo	Locally runnable Streamlit web application.
Final report	Academic report describing the limitations of text-only pipelines, the multimodal API measured benchmark results.